Learning from and about climate scientists

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Abstract

Despite the overwhelming scientific consensus that human activities contribute significantly to climate change, public opinion remains divided. To bridge this gap, informative messaging about the consensus has been widely proposed as a persuasive tool. However, it remains challenging to understand how people interpret this information and how it interacts with their wider belief system. Using survey experiments that vary the level of scientific consensus, we find that consensus information not only influences climate change beliefs but also shapes perceptions of climate scientists themselves, consistent with normative principles of Bayesian belief updating. Notably, climate beliefs are strongly linked to perceptions of scientist skill which highlights a potential avenue for communication. By unpacking the belief system underlying one of the most prominent climate communication strategies, our research provides a deeper understanding of public response to consensus messaging, offering guidance for developing more targeted and effective science communication.

Keywords: Climate communication | Expert consensus | Source credibility | Bayesian inference

While nearly all climate scientists agree that human activities play a central role in climate change (Oreskes, 2004; Anderegg et al, 2010; Cook et al, 2013; Cook and Lewandowsky, 2016; Myers et al, 2021), most Americans underestimate this consensus (Leiserowitz et al, 2021). Individuals who think the consensus is high tend to believe more in human-caused climate change (Lewandowsky et al, 2013; Hornsey et al, 2016; Bertoldo et al, 2019) and support policies aimed at mitigating it (Ding et al, 2011; McCright et al, 2013). Accordingly, one of the most prominent methods for persuading people of human-caused climate change involves informing them that 97% of climate scientists believe it is happening (van der Linden et al, 2015, 2019; Cook et al, 2018; van der Linden, 2021; Rode et al, 2021; Deryugina and Shurchkov, 2016). However, while climate consensus messaging has received considerable attention, it remains challenging to understand how exactly people interpret this information and how it interacts with their wider belief systems (Cook and Lewandowsky, 2016; Dieckmann et al, 2017; Johnson and Dieckmann, 2018; Dieckmann and Johnson, 2019; Gustafson and Rice, 2020; Landrum and Slater, 2020; Bayes et al, 2020).

Perceptions of scientist credibility play a key role in the public's responsiveness to science communication (Rabinovich et al, 2012; Lupia, 2013; Fiske and Dupree, 2014; Cologna and Siegrist, 2020). Suppose you learn that 97% of climate scientists agree that human activities are causing substantial changes to the climate. Is this consensus so strong because climate scientists are knowledgeable and unbiased experts who have converged upon the truth? Or is it strong because they form a biased and conspiratorial cabal? The weight placed on each possibility will determine whether one believes the consensus position. Perceived credibility may affect not only the interpretation of a message, but also receptiveness to subsequent messaging, willingness to seek out information from scientific sources (Zhang, 2023), and support for funding of science (Ophir et al, 2023). Such perceptions have been tied to other elements of personal worldview like political orientation (Zhang et al, 2018). For example, Democrats are more likely than Republicans to believe that climate scientists understand climate change very well and are primarily driven by evidence and public interest (pew, 2016). These attitudes can be quite enduring (Motta, 2018). Consequently, expert credibility is one of the most hotly contested targets in the battle for public opinion over climate science (Hmielowski et al, 2014; Cook et al, 2018) and far beyond (Nichols, 2017).

In this study, we experimentally examine the nuanced interplay between consensus messaging, source credibility, and political orientation, offering a more comprehensive and holistic view of how people interpret and respond to scientific consensus on human-caused climate change. We leverage a normative Bayesian belief updating perspective to delve more deeply into how people update their beliefs about climate change and climate scientists based on varying levels of the consensus.

While it is tempting to view climate beliefs through the lens of irrationality, these beliefs may nonetheless invoke rational reasoning processes (Cook and Lewandowsky, 2016; Druckman and McGrath, 2019; Bago et al, 2023). Critically, we can only understand these beliefs by specifying how the reasoning process operates not just on information but on the information channel itself. Normative principles of belief updating offer a versatile and generalizable framework for investigating belief systems. This lens enables us to systematically link key variables like consensus information, source credibility, subjective interpretation, and belief change (Hahn, 2020; Oaksford and Chater, 2020; Shafto et al, 2012; Gershman, 2019; Bhui and Gershman, 2020). For this reason, such principles are suspected to reveal important qualitative properties of public response to scientific communication on climate change (Cook and Lewandowsky, 2016; Hahn et al, 2016; Druckman and McGrath, 2019) and enhance our ability to predict belief updating (Powell et al, 2023).

These principles imply that the way a person responds to data—such as the content of a message—depends on how they attribute those data to possible underlying causes. Whether one responds to scientific consensus should depend on whether they think scientists are credible. More subtly, beliefs about the underlying causes may themselves be simultaneously shaped by the data in a reciprocal manner (Bovens and Hartmann, 2003; Landrum et al, 2015; Hahn et al, 2016; Perfors et al, 2018; Madsen et al, 2020; Merdes et al, 2021). For instance, if people are uncertain about how credible climate scientists are in general, the consensus itself can shape their perceptions of these traits, as illustrated in Figure 1. A field in wide disagreement about concrete facts can hardly be uniformly biased, nor can the scientists working in that field all be reliably capable of discerning the truth. Judgments may moreover be richly contingent on specific qualifiers, such as when perceptions of credibility depend on the stated position of an individual scientist (Anderegg et al, 2010). In short, when provided with consensus information, people not only learn *from* scientists, they may also learn *about* scientists. While implications like these have been corroborated in more abstract settings, they have seen limited investigation in the empirical landscape of climate communication (cf. Cook and Lewandowsky, 2016).

We explore how people interpret information about scientific consensus on human-caused climate change, using survey experiments. In a U.S. nationally-representative sample (N = 1,737) and two convenience samples (see SI), we investigate how consensus messaging affects beliefs about *scientists* alongside beliefs about climate change. We focus on perceptions of climate scientists along two key dimensions which are commonly considered to comprise source credibility (Pornpitakpan, 2004; Lupia, 2013; Fiske and Dupree, 2014): skill and bias (aka expertise and trustworthiness). To probe the effects of consensus as directly as possible, our experiments vary the hypothetical level of agreement (e.g., from 50% through 99%; Aklin and Urpelainen, 2014; Kerr and Wilson, 2018; Chinn et al, 2018; Johnson, 2018), and elicit beliefs about the skill and bias of scientists who agree or disagree with the consensus. We further study the effects of two interventions focused on climate scientist credibility.

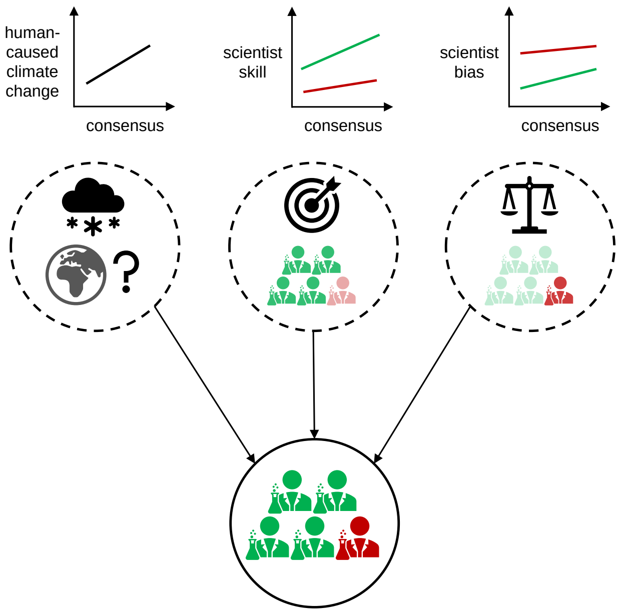

Fig. 1: **Illustration of Hierarchical Inference.** Inference about the consensus among scientists (bottom bubble) requires making judgments about its latent causes, such as the true state of the world (left bubble), the ability of scientists to detect this state (middle bubble), and their bias in reporting an opinion (right bubble). The consensus provides information about scientists collectively and individually based on expressed pro-consensus (green) or anti-consensus (red) opinion.

We find evidence that people draw sophisticated inferences *about* scientists in addition to learning from them. First, greater consensus raises average belief in human-caused climate change for all political orientations, consistent with persuasion often occurring in parallel (Coppock, 2022) and the documented effectiveness of the consensus message (van der Linden et al, 2018). Second, evaluations of scientist skill and bias are contingent on overall consensus level and on the scientist's individually expressed (pro- or anti-consensus) position. These patterns of complex inference vary by political party. Third, consensus affects climate belief in part via perceptions of scientists, especially the perceived skill of pro-consensus scientists. Fourth, the ways in which people learn about scientists fall into clusters, characterized partly by whether consensus predominantly affects perceptions of skill or of bias, and linked to political orientation. Fifth, the degree to which consensus affects a person's beliefs about climate change is correlated with the degree to which it affects their beliefs about climate scientists, in a manner that varies across the clusters. Finally, informing people about (i) steps taken by scientists to reduce bias and (ii) the lengthy history of climate science both increase perceived pro-consensus scientist skill, but only the latter increases belief in human-caused climate change (over and above the effect of consensus).

From a theoretical perspective, our research demonstrates the complex interdependence of beliefs about climate change and perceptions of climate scientists, highlighting the sophisticated manner in which individuals interpret consensus information. The data support the conception of an intricate, conditional belief structure about scientists, as well as the hypothesis that people are updating this schema based on consensus information. From an applied perspective, the data suggest that perceptions of pro-consensus scientist skill are especially malleable and linked to climate beliefs, raising an important mediating variable. Overall, our research helps methodically unpack one of the most influential climate communication strategies by mapping out the belief system upon which it rests, offering valuable insight for science communication.

1 Results

1.1 Learning from scientists: the effect of consensus on climate belief

In the following sections, we focus on mapping the belief system using the control arm of a nationally representative sample collected on Bovitz.[1] We first observe that belief in human-caused climate change is influenced by scientific consensus. Figure 2 demonstrates that, on average and across political parties, subjects revise their beliefs to reflect a higher likelihood that human-caused climate change is occurring as consensus among climate scientists increases. Across the sample, each percentage point increase in consensus led to a 0.29pp increase in climate belief on the 0–100 scale ($p < 0.01$; $\beta_{Democrat} = 0.35$, $\beta_{Independent} = 0.20$, $\beta_{Republican} = 0.29$; see Table S3 in the SI). Sensitivity to consensus appears correlated with prior beliefs, as those with higher initial perceptions of scientist skill respond more to consensus (Table S4).

1.2 Learning about scientists: the effect of consensus on perceptions of scientists

A key hypothesis that distinguishes the normative Bayesian framework of joint inference is the prediction that, in addition to drawing inferences about climate change, consensus messaging will lead people to draw inferences about the characteristics of climate scientists. To test this, we asked participants how likely they thought a randomly selected climate scientist who expressed belief or disbelief in human-caused climate change ("pro-consensus" and "anti-consensus" scientists respectively) would be biased (i.e., would always express the same opinion regardless of the evidence), as well as how likely an unbiased

[1]There are two additional arms—which receive information about climate scientists in conjunction with the consensus message—that are explored in a later section in comparison to this control group.

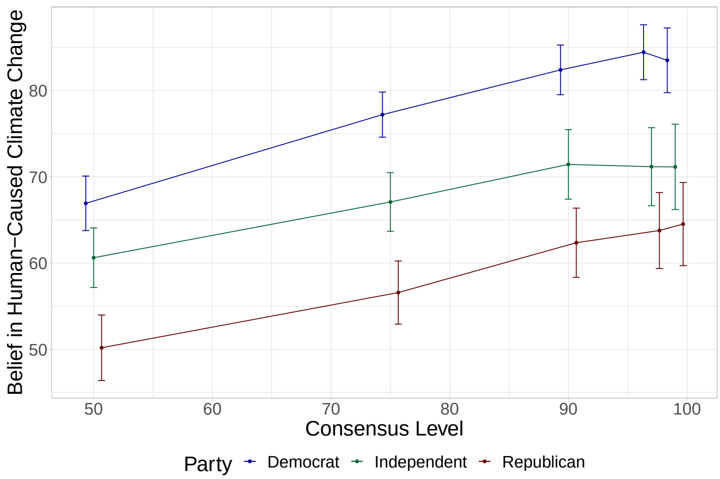

Fig. 2: **The Responsiveness of Belief in Human-Caused Climate Change to Scientific Consensus, Split by Political Party.** The mean belief is plotted at 50%, 75%, 90%, 97%, and 99% consensus with 95% confidence intervals.

scientist who expressed each position would be skilled (i.e., arrived at their conclusion due to the evidence). These queries were posed for each consensus level.

As Figure 3 shows, participants hold substantially differentiated perceptions of scientists based on those scientists' expressed pro- or anti-consensus position (see Table S2). Comparing among political parties, Democrats consider pro-consensus scientists to be more skilled and less biased than anti-consensus scientists, while Republicans hold the opposite view, and Independents fall in the middle (see Table S2).

Looking across political parties, belief in the skill and bias of pro- and anti-consensus scientists generally vary with consensus level. On average, a one percentage point increase in consensus raises perceptions of pro-consensus scientist skill by 0.17pp ($p < 0.01$), anti-consensus scientist skill by 0.12pp ($p < 0.01$), pro-consensus scientist bias by 0.11pp ($p < 0.01$), and anti-consensus scientist bias by 0.11pp ($p < 0.01$). See Table S3 for regressions interacting consensus level with political party.

It may appear counter-intuitive for the skill and bias of both types of climate scientists to rise together with consensus. However, this pattern of belief updating is consistent with a process of joint inference. In particular, we propose a hierarchical Bayesian model (detailed in Section S6 of the SI) where the message recipient makes joint inferences over three types of beliefs when presented with evidence (i.e., consensus): 1) the focal hypothesis that human-caused climate change is occurring, 2) the attributes of individual climate scientists, and 3) the general attributes of the field. Intuitively, if there is some

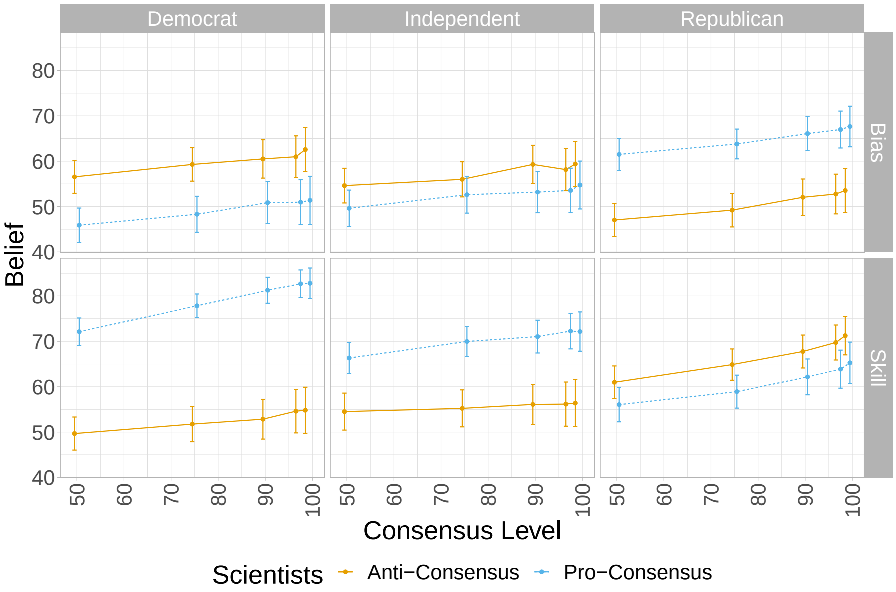

Fig. 3: **The Responsiveness of Beliefs About the Attributes of Climate Scientists to Scientific Consensus, Split by Political Party.** The mean belief is plotted at 50%, 75%, 90%, 97%, and 99% consensus with 95% confidence intervals.

commonality between the traits of various scientists, due for instance to the difficulty of the field, then perceptions of them may move in tandem when information is obtained.

Similarly, we also found that consensus affects perceptions of climate scientists in two other experiments using non-representative convenience samples from Amazon Mechanical Turk (described in Section S2 of the SI). The first of these uses a comparable within-subject design. The second experiment uses a between-subject design in which each participant is only shown a single consensus level, and participants are not asked directly about their climate belief, to alleviate the possibility of spillover effects across different consensus levels. In the within-subject replication, we find almost identical patterns of belief updating with the exception that Democrats decrease their belief in pro-consensus bias and do not increase their belief in pro-consensus skill. In the between-subject version, we find that the slopes of pro-consensus bias and anti-consensus skill are negative, while the other belief updates are positive but statistically insignificant. These differences should not be surprising as belief patterns emerge in different manners for subgroups with similar findings for heavily Democratic populations, as shown in Section 1.4, and the Mechanical Turk samples skew heavily Democratic.

1.3 Measuring the strength of direct and indirect effects

While we suggest that a process of joint inference is occurring where participants simultaneously update their beliefs about climate change and climate scientists, we use a structural equation model (SEM) to decompose the measured belief system in a more theory-agnostic manner. We analyze perceptions of scientist credibility as mediators of consensus effects (Chinn et al, 2018), expanding credibility into the four metrics we measure.

Figure 4 demonstrates a direct relationship between scientific consensus and climate change; belief in human-caused climate change grows by 0.17pp ($p < 0.01$) for each percentage point of scientific consensus. Additionally, there are strong, statistically significant relationships between scientific consensus and beliefs about the attributes of climate scientists as discussed before. Finally, higher perceptions of pro-consensus scientist skill or anti-consensus scientist bias are linked to greater climate belief while anti-consensus scientist skill is associated with lower climate belief. These three results respectively reflect learning from scientists, learning about scientists, and the indirect impact of shifting perceptions of scientists. Similar results are obtained from the Amazon Mechanical Turk sample (Figure S3). Considering the first and second stages together, the indirect "pro-consensus skill" pathway (consensus $\rightarrow$ perceived pro-consensus skill $\rightarrow$ climate belief) has an effect that is 62% as strong as the direct pathway, indicating a sizeable impact.

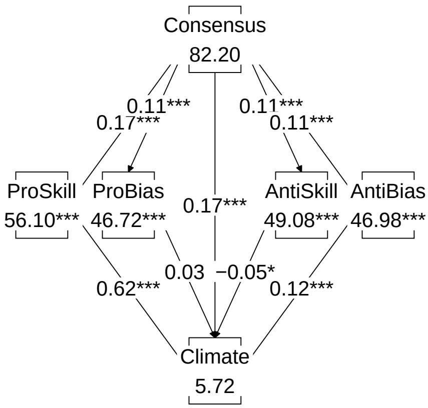

Fig. 4: **Structural Equation Model Demonstrating Direct and Indirect Pathways between Consensus and Belief in Human-Caused Climate Change.** Beliefs are expressed on 0-100 scales and standard errors are clustered by individual. Consensus is the provided consensus level among scientists, ProSkill and ProBias are belief in the skill and bias of pro-consensus scientists while AntiSkill and AntiBias are the same for beliefs for anti-consensus scientists.

1.4 Patterns of belief updating reveal distinct demographic clusters

Given that ideology and worldview are particularly strong predictors of climate belief (Hornsey et al, 2016), we next explore heterogeneity in our measured belief systems through cluster analysis. Identifying distinct clusters can enable us to pinpoint specific characteristics and tendencies that influence how certain groups revise their beliefs, ultimately allowing for more targeted and effective communication strategies. By examining the resulting clusters, we can determine whether they closely correspond to variables like political orientation or if they transcend political boundaries.

Using k-means clustering, we categorize individuals into four distinct groups based on the degree to which their perceptions of scientists are influenced by consensus levels. This approach enables us to identify specific patterns of belief updating that turn out to correspond in part to political orientation. The clusters are presented in descending order by cluster size where Cluster 1 contains 236 participants (40%), Cluster 2 contains 152 participants (25%), Cluster 3 contains 123 participants (21%), and Cluster 4 contains 86 participants (14%).

Figure 5A illustrates the ideological and demographic characteristics of each group. Cluster 1 is disproportionately Republican and characterized by low prior belief in climate change, consensus, and the skill of climate scientists, as well as higher prior belief in the bias of climate scientists. Cluster 2, by contrast, is predominantly Democratic and characterized by prior beliefs more inclined towards accepting the existence of human-caused climate change. Cluster 3 also leans Republican while cluster 4 has an intermediate balance. Thus, even though the clustering was based solely on patterns of updating beliefs about scientists, the resulting groups do reflect political party.

Figure 5B demonstrates how climate beliefs in each cluster are affected influenced by scientific consensus. Clusters 1 and 2 are both persuaded by consensus with Cluster 1 having lower belief in climate change at each level, reflecting their lower priors. Cluster 3 exhibits strong positive updating while Cluster 4 exhibits negative changes. In sum, 3 of the 4 clusters are persuaded by consensus with even Cluster 1—the most Republican and a-priori skeptical group—demonstrating an approximately 15pp increase in climate belief over the range of consensus levels.

Figure 5D shows that as consensus on human-caused climate change increases among scientists, individuals in Cluster 1 perceive those scientists to be more skilled. However, their perceptions of bias hardly change. Conversely, individuals in Cluster 2 show an increase in perceived bias but negligibly update perceptions of skill (which are already near-ceiling for pro-consensus scientists). Cluster 3 and 4 respectively exhibit strong positive and negative changes across the board. While the patterns from the first three clusters are more naturally accommodated by the Bayesian model, the fourth (and smallest) cluster does not seem to match any theoretical patterns.

Figure 5C displays correlation matrices between the slopes of these belief curves, capturing the interconnected nature of the belief systems. Climate belief updating in Cluster 1 is positively associated with perceived skill updating, particularly pro-consensus scientist skill. A negative correlation is found between anti-consensus scientist bias updating and skill. Climate belief updating in Cluster 2 is positively correlated with pro-consensus skill and bias updating, while updating of anti-consensus bias and anti-consensus skill are negatively associated. Climate belief updating in Cluster 3 is positively associated with perceived skill updating. All curves in Cluster 4 are generally positively correlated with each other.

The clusters thus appear to recapitulate political orientation to a substantial extent. The analysis reveals a largely Republican group with more malleable perceptions of climate scientist skill, and a largely Democrat group with more malleable perceptions of climate scientist bias. Moreover, updating of climate beliefs is consistently associated with updating of perceived pro-consensus scientist skill. Taken together, these findings suggest that perceptions of skill may be especially important and more amenable to change.

1.5 Intervening on skill and bias

We explore two interventions to change perceptions of scientific credibility in tandem with consensus information (see SI S4). Each comprised a paragraph of information which was provided to participants in the treatment arms in addition to consensus level. The first intervention—the "History intervention"—describes the lengthy history of climate science, while the second—the "Institutions intervention"—describes the institutional constraints on scientists to increase credibility. The text of each intervention is presented in Section 3.6. Note that we do not claim these are the sole, prototypical, or best interventions dealing with credibility.

The "History" intervention appears to be particularly effective. Controlling for consensus level, the history intervention leads to a 3.90pp ($p < 0.001$) increase in belief in human-cause climate change in a regression of belief on intervention-type. At 50%, 75%, and 90% consensus levels, the history intervention leads to significant increases in belief in human-caused climate change of 5.5pp ($p < 0.001$), 4.5pp ($p < 0.01$), and 3.7pp ($p = 0.02$), respectively when regressing belief on a dummy for intervention-type. At the higher consensus levels, the effect is positive but insignificant as many participants are already near the ceiling. Thus it may be effective when consensus is underestimated. Meanwhile, the "Institutions" intervention is insignificant at each consensus level with substantially smaller effect sizes ranging between 1.6pp and 2.5pp.

Figure 6 shows the average effect of each intervention on belief in human-caused climate change and perceptions of scientist credibility. In both interventions, the effects on bias are not significantly different from zero, indicating that beliefs about bias are particularly difficult to shift. For both interventions, there is an approximately 3pp increase in perception of the skill of pro-consensus scientists. However, there appears to be a larger decrease

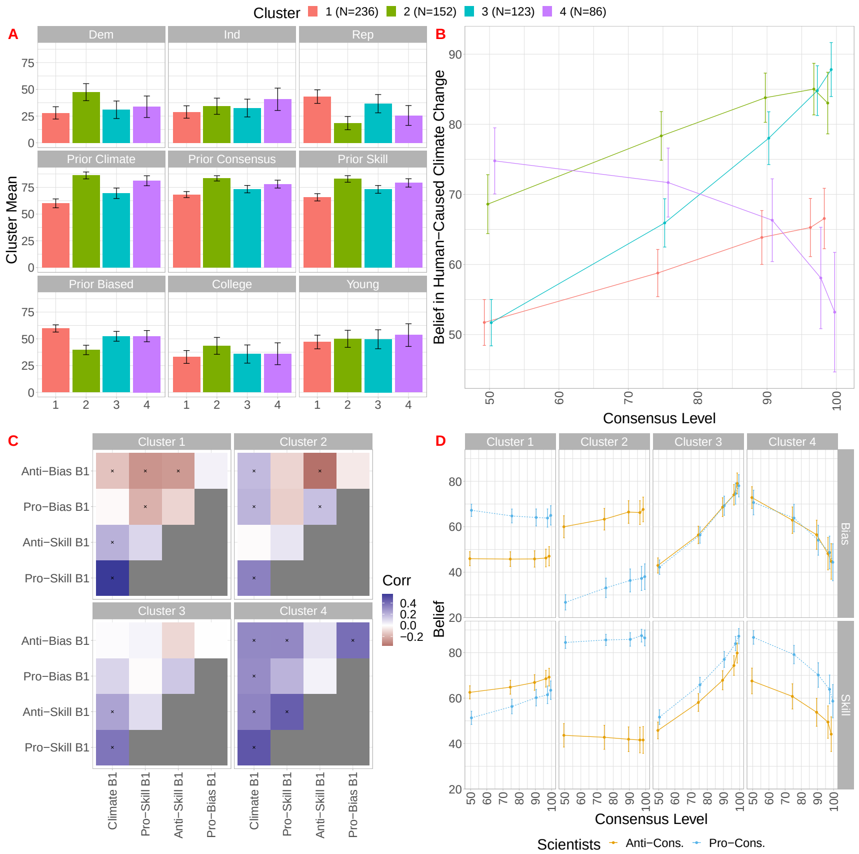

Fig. 5: **The Population can be Decomposed into 4 Distinct Clusters that have Different Belief Patterns and Correspond to Different Demographics and Prior Beliefs.** Panel A shows the percent of the cluster that is Democrat, Independent, and Republican, attended college, has a below median age (Young), and their corresponding average prior beliefs. Panel B shows the average belief in human-caused climate change at each consensus level split by cluster. Panel C shows the correlation between the belief slopes within each cluster. Statistically significant correlations are marked with an "x". Finally, Panel D shows the average belief in each of the 4 scientist attributes split by cluster. 95% confidence intervals are shown around all means.

(approximately 2x) in beliefs about the skill of anti-consensus scientists in the history intervention compared to the institutions intervention.

Taken together, it appears that the credibility of climate scientists is linked to belief in human-caused climate change and that perception of skill is the more malleable component of credibility.

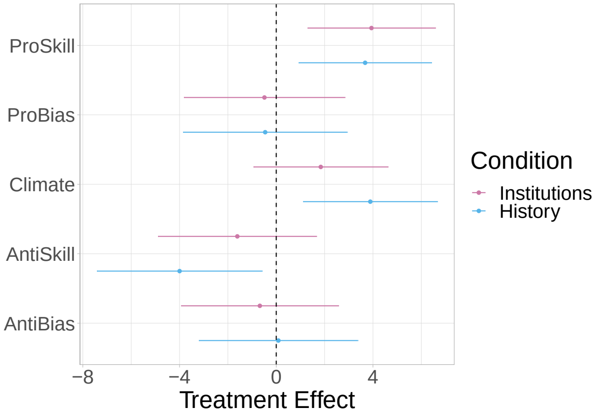

Fig. 6: **Treatment Effect of the History and Institutions Interventions on Beliefs about Climate and Scientist Attributes.** Treatment effects are plotted from a regression of the focal belief on a dummy for each experimental condition with controls for each consensus level. 95% confidence intervals are shown around all treatment effects from individual-clustered standard errors.

2 Discussion

Perceived scientist credibility plays a crucial role in the public's reception of science communication, particularly for politically charged topics such as climate change (Cook et al, 2018; Lupia, 2013; Fiske and Dupree, 2014). The battle for public opinion often hinges on trust in the scientific community, a factor that can be significantly influenced by external forces like partisan media (Hmielowski et al, 2014) or by the actions of scientific organizations themselves (Zhang, 2023). In this context, understanding the interplay between consensus messaging, source credibility, and political orientation is essential for devising effective communication strategies that can bridge the gap between public perception and scientific understanding.

We experimentally investigate this complex relationship, providing novel insights into how people interpret and respond to scientific consensus on human-caused climate change. Our research uncovers the interconnected nature of beliefs about climate change and perceptions of climate scientists, revealing the sophisticated manner in which individuals process consensus information. The results point to an intricate conception of trust in scientists which is contingent on overall consensus level and on a scientist's individually expressed position, as implied by normative principles of Bayesian inference. Furthermore, consensus affects climate belief not only through a direct channel but also indirectly via beliefs about scientists. Patterns of learning about scientists fall into clusters characterized partly by whether consensus predominantly affects perceptions of skill or of bias, and which were linked to political orientation.

The largest cluster of respondents, who initially hold the most doubt about human-caused climate change and climate scientists, exhibit a higher sensitivity to consensus in terms of perceived scientist skill. Moreover, the extent to which people update their beliefs about pro-consensus scientist skill is consistently correlated with changes in their beliefs about human-caused climate change. Notably, the indirect effect of consensus on climate change belief is most strongly modulated by perceptions of pro-consensus scientist skill. These findings suggest that communication strategies emphasizing the expertise of climate scientists could prove especially effective in changing beliefs.

More broadly, expertise might be a fruitful avenue of intervention. Expertise is often seen as an objective and quantifiable attribute, which can be demonstrated through credentials, experience, and a proven track record of accurate predictions. Emphasizing this aspect can help build trust in expert opinions, as it highlights the depth of knowledge and competence that underlies their conclusions. Much of the science communications literature has focused on increasing the trustworthiness of scientists but our results in the first experiment suggest that intervening on perceptions of skill can provide an alternative and perhaps more malleable route to increasing scientist credibility.

We leverage the belief network unpacked in the first experiment to design two interventions focused on shifting perceptions of climate scientist trustworthiness and skill. Providing messages about the credibility of climate scientists can increase belief in human-caused climate change on top of consensus information. Furthermore, perceptions of the skill of climate scientists appear particularly malleable, suggesting a route through which perceptions of scientist credibility may be influenced. Beyond belief in human-caused climate change, which we focus on given its prominence in the consensus messaging literature (van der Linden et al, 2015; van der Linden, 2021), our framework may connect to other relevant judgments like support for sustainability policies or willingness to undertake individual action.

The Bayesian framework which motivates us offers hope of precisely characterizing which beliefs will most resist change in the face of information (Botvinik-Nezer et al, 2023), as well as which pieces of information will be most persuasive for which people (Powell et al, 2023). Tailored messaging strategies that account for individual differences in responses to consensus messaging might be developed to more effectively resonate with diverse audiences. As models like these have also been applied to capture the sophisticated manner in which even children learn about source credibility (Bonawitz and Shafto, 2016), they may help characterize how trust in scientists develops and crystallizes over the life cycle (Motta, 2018).

Finally, the approach we take can be generalized to other domains where learning from experts is essential, such as health, economics, and technology. In these fields, the credibility of experts also plays a vital role in shaping public opinion and informing policy decisions (Algan et al, 2021; pew, 2019). While perceptions of experts will vary across domains, the underlying process of interpretation may follow similar principles. Establishing credibility in reliable sources of knowledge can lead to a more informed public discourse, increased acceptance of evidence-based policies, and ultimately, better outcomes in crucial areas such as public health, economic stability, and technological advancement.

3 Methods

3.1 Participants

Our primary sample consists of 1,737 participants from a nationally representative American sample recruited on Bovitz – a high-quality recruitment platform that specializes in representative samples (for a paper demonstrating the benefits of Bovitz in data collection see Voelkel et al, 2023). A summary of this sample can be see in Table S6 in the SI. The sample closely matches the US population more broadly, especially on partisanship.

The second sample, which received the same survey design and is used for robustness checks, included 493 participants recruited on Amazon Mechanical Turk (MTurk). The demographics of this sample are shown in Table S11. The sample skews heavily Democrat (49%), younger, and White (83%) as is expected on MTurk.

Finally, the last sample, which received a between-subject survey design, includes 432 participants that were also recruited from MTurk. As with the second sample, the participants skew heavily Democrat (51%), younger, and White (78%) as shown in Table S12.

Participants provided informed consent, and the studies were approved by the MIT Committee on the Use of Human as Experimental Subjects (COUHES).

3.2 Design

In the primary study design, participants completed a survey composed primarily of questions concerning demographics, their beliefs about climate change, and their beliefs about climate scientists. In the demographics section, we collected data on traditional demographics (age, race, gender, income, and education) plus political demographics (party affiliation, political extremity, and affective polarization). The composition of the sample is shown in Table S6 and skews white, educated, and Democratic relative to the country.

Next, we asked participants their belief in human-caused climate-change, their belief in the consensus level among climate scientists, and their confidence in each of these estimates. Additionally, we asked the likelihood that a climate scientist is extremely biased, meaning they will ignore the evidence in their evaluation of whether climate change is happening; and the likelihood that an extremely biased scientist will be biased to say climate change is or is not occurring. The product of these two beliefs gives the prior probability that any given climate scientist is biased in a certain direction. Lastly, we asked participants the likelihood that, conditional on climate change (not) occurring, an unbiased climate scientist will correctly identify this using their skill.

Finally, we presented participants with the main questions of interest relating to their beliefs about human-caused climate change and pro-consensus scientists (those who express that climate change is occurring) and anti-consensus scientists (those who do not). For each belief—human-caused climate change, the skill of pro-consensus scientists, the skill of anti-consensus scientists, the bias of pro-consensus scientists, the bias of anti-consensus scientists—we asked participants to hypothetically consider 5 different levels of consensus among climate scientists [50%, 75%, 90%, 97%, 99%] and report their belief at that consensus level using a 0–100 scale.

In the between-subjects design, participants completed a similar set of demographic questions. They were then asked to consider a single consensus level, randomly assigned to be 50%, 75%, or 97%. For this consensus level, they answered only the questions about the attributes of climate scientists.

3.3 Belief Sensitivity

To quantify the effect of consensus on a given belief, we regress that belief on a continuous variable for consensus, normalizing it to start at 50. This regression can be expressed:

$$y_{ic} = \beta_0 + \beta_1 consensus_c + \epsilon_{ic} \tag{1}$$

where y_{ic} is individual i's belief in hypothesis y at consensus level c. β_1 gives a linear approximation of the effect of consensus on that belief. To estimate the population-level sensitivity of climate and perceived scientist attribute beliefs to consensus, we conducted this regression across all individuals in the sample with standard errors clustered for the individual.

To estimate individual sensitivity to consensus, we used the same regression restricting to only observations for participant i. With this setup, β_0 gives an estimate for that individual's belief in the hypothesis at 50% consensus and β_1 gives the average sensitivity of that belief to consensus.

3.4 Structural Equation Models and Reduced Form Equivalents

In Figure 4, we present the results of a structural equation model where the effect of consensus on climate change is moderated by scientist attributes. The model can be summarised by 5 first-stage equations with consensus as the independent variable predicting the dependent belief variables of human-caused climate change (Climate), the skill and bias of pro-consensus scientists—ProSkill and ProBias—and the skill and bias of anti-consensus scientists—AntiSkill and AntiBias. The second stage is summarized by 4 equations where the 4 scientist attributes predict belief in human-caused climate change. For all equations, standard errors are clustered at the individual-level. Figure S3 performs the same analysis on the within-subject MTurk sample.

3.5 Clustering

To produce the clusters presented in Section 1.4, we use k-means clustering with 4 centers. We chose 4 centers to balance interpretability and differentiation of the clusters. The qualitative results are similar with 3 or 5 clusters.

3.6 Interventions

The text of the interventions is presented below:

- *Institutions*: Scientists and universities take many steps to reduce systemic bias and make their science objective and open. Scientific journals require conflict of interest statements and other academics, journalists, and public interest organizations investigate the funding sources of scientists. Those who receive funding from vested interests are often sanctioned by their peers and their research is taken less seriously. Scientists want their papers to be available to the public and generate conversation about important topics, with a survey showing that over 95% of scientists think public access to their research is important.
- *History*: Climate change might seem like a sudden and modern problem, but the study of climate change is actually a long-established science. The basic dynamics of carbon dioxide (C02) and atmospheric warming were first understood in 1860 by physicist John Tyndall, based on earlier insights by Joseph Fourier in 1824. To track climate, scientists have been recording temperature measurements for hundreds of years, and CO2 has been measured directly at Mauna Loa Observatory since 1958. Famously, NASA scientist James Hansen testified to Congress in 1988 stating the greenhouse effect had been detected.

To estimate the average effect of the interventions, we regressed belief in human-caused climate change at each consensus level on a dummy for treatment assignment and a continuous control for consensus level. Standard errors are clustered by respondent. Meanwhile to estimate the effect of the intervention at each consensus level, we regress belief in human-caused climate change on a dummy variable for treatment assignment restricting to that consensus level. Standard errors are robust to heteroskedasticity.

3.7 Acknowledgments

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S1 Supplementary Information I: Main Bovitz Sample

In this section of the Supplementary Information, we provide additional analyses and robustness checks for the primary sample. The order is arranged in the same manner as the main paper with additional information about the sample in the final subsection.

S1.1 Learning from Scientists

Table S1 summarizes the individual-level belief sensitivity estimates from Equation 1. The mean and standard deviation describe the distributions of β_1 over the population while $Pct > 0$ expresses the percent of β_1's in the population that are greater than 0. The t-statistic is from a two-sided t-test of the distribution of β_1's against the null-hypothesis that the mean is 0.

Table S1: Summary Statistics of Individual-Level Sensitivity to Consensus

Variable	N	Mean	SD	Pct ¿ 0	T-stat
Climate Slope	597	0.29	0.69	0.73	t = 10.31
Pro-Skill Slope	597	0.17	0.55	0.70	t = 7.72
Anti-Skill Slope	597	0.12	0.58	0.64	t = 4.74
Pro-Bias Slope	597	0.11	0.63	0.62	t = 4.26
Anti-Bias Slope	597	0.11	0.64	0.63	t = 4.28

Notes: The mean and standard deviation of the distribution of sensitivity to consensus is shown for each belief type. Sensitivity to consensus is measured using an individual-level linear regression of belief on consensus level. The column titled $Pct > 0$ provides the proportion of the distribution that has a positive slope. Finally, a t-stat from a t-test of each distribution against the null-hypothesis that the mean of the distribution is 0 is presented.

To measure baseline differences in beliefs about climate scientists between scientist types and political party, we use a linear regression of belief in a focal hypothesis (the skill or bias of a scientist group) on a dummy for whether the scientist is pro-consensus and, in one version, its interaction with political party. In the first version, we include fixed-effects for individual and consensus level to control for variation between participants and across consensus. The first regression can be expressed:

$$y_{ijc} = \gamma_i + \zeta_c + \beta_1 Pro_j + \epsilon_{ijc} \tag{2}$$

where y_{ijc} is individual i's belief in scientist type j's attribute y (skill or bias) at consensus level c. γ_i and ζ_c are individual and consensus fixed-effects, and β_1 gives the baseline (residualized) difference between pro- and anti-consensus scientists. A second version of the regression, which tests differences in these beliefs by political party replace γ_i with dummy variables for political party identification and their interactions with Pro. Standard errors are clustered at the individual-level for each of these regressions. Table S2 presents these results for differences by political party.

Table S2: Regression of Individual Beliefs on Scientist Type and Party Identification

Dependent Variables:	Skill		Bias	
Model:	(1)	(2)	(3)	(4)
Variables				
Pro	12.06***	-5.670**	-0.4395	14.28***
	(1.547)	(2.627)	(1.651)	(2.514)
Dem		-14.18***		9.058***
		(2.711)		(2.728)
Ind		-11.24***		6.568**
		(2.756)		(2.764)
Pro × Dem		32.27***		-24.78***
		(3.552)		(3.784)
Pro × Ind		20.34***		-19.03***
		(3.629)		(3.850)
Fixed-effects				
ID	Yes	No	Yes	No
Consensus	Yes	Yes	Yes	Yes
Fit statistics				
Observations	5,970	5,970	5,970	5,970
R^2	0.44244	0.10045	0.41626	0.03503
Within R^2	0.06820	0.09345	8.41×10^{-5}	0.03095

Clustered (ID) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: This table presents four regressions using the form presented in Equation 2. Columns (1) and (2) regress an individual's belief in the skill of scientists at a given consensus level on a dummy for if the scientist is pro-consensus and, in Column (2), dummies for political party identification and their interaction with *Pro*. Columns (3) and (4) use the same models with scientist bias as the dependent variable. In all columns, there are consensus-level fixed-effects while columns (1) and (3) also include individual fixed-effects. In all models, standard errors are clustered by individual.

To test for differences in sensitivity to consensus by political party identification, demographics, and prior beliefs, we regress the individual-level sensitivity on the covariates of interest. The regression can be expressed:

$$sensitivity_i = \beta_0 + X_i\beta + \epsilon_i \tag{3}$$

where $sensitivity_i$ is individual i's sensitivity of the belief of interest to consensus as measured by Equation 1 and X_i is a matrix of individual-level predictors. Tables S3 and S4 use this form of analysis.

Table S3: Regression of Individual Sensitivity of Beliefs to Consensus Level on Party Identification

Dependent Variables:	Climate	ProSkill	AntiSkill	ProBias	AntiBias
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Constant	0.2246***	0.1188***	0.0382	0.0913**	0.0961**
	(0.0497)	(0.0395)	(0.0415)	(0.0452)	(0.0458)
Dem	0.1294*	0.1027*	0.0641	0.0229	0.0117
	(0.0694)	(0.0552)	(0.0580)	(0.0631)	(0.0639)
Rep	0.0727	0.0614	0.1600***	0.0320	0.0355
	(0.0701)	(0.0557)	(0.0586)	(0.0637)	(0.0646)
Fit statistics					
Observations	597	597	597	597	597
R^2	0.00583	0.00586	0.01257	0.00045	0.00053
Adjusted R^2	0.00249	0.00251	0.00925	-0.00291	-0.00283

IID standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: This table presents five regressions using the form presented in Equation 3. In each model, an individual's sensitivity of a belief to consensus, captured by the slope of the individual-level regression of the belief on consensus, is regressed on a constant and their party identification. The dependent variables are, in order, belief in human-caused climate change, the skill of pro-consensus scientists, the skill of anti-consensus scientists, the bias of pro-consensus scientists, and the bias of anti-consensus scientists. Classical standard errors are used in each model.

Table S4: Regression of Individual Sensitivity of Beliefs to Consensus Level on Demographics

Dependent Variables:	Climate		ProSkill		AntiSkill		ProBias		AntiBias	
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Variables										
Constant	0.2213**	0.0618	0.1116	0.0685	-0.0285	0.2131***	-0.0382	0.1914**	-0.0481	0.2258**
	(0.0932)	(0.0960)	(0.0742)	(0.0766)	(0.0781)	(0.0809)	(0.0844)	(0.0874)	(0.0857)	(0.0886)
Age	-0.0015		0.0001		0.0019		0.0012		0.0012	
	(0.0018)		(0.0014)		(0.0015)		(0.0016)		(0.0016)	
Female	0.0669		-0.0278		-0.0186		-0.0128		0.0351	
	(0.0571)		(0.0455)		(0.0479)		(0.0517)		(0.0525)	
College	-0.0028		0.0012		-0.0412		-0.0057		0.0597	
	(0.0600)		(0.0478)		(0.0503)		(0.0543)		(0.0551)	
White	0.1495**		0.0935*		0.0990*		0.1338**		0.0829	
	(0.0689)		(0.0549)		(0.0578)		(0.0624)		(0.0634)	
Prior(Climate)		-0.0010		-0.0010		-0.0017		-0.0009		-0.0017
		(0.0012)		(0.0010)		(0.0010)		(0.0011)		(0.0011)
Prior(Unbiased)		0.0018		0.0011		-0.0004		-0.0014		0.0008
		(0.0011)		(0.0009)		(0.0010)		(0.0010)		(0.0011)
Prior(Skill)		0.0029**		0.0017		0.0006		0.0007		-0.0004
		(0.0015)		(0.0012)		(0.0013)		(0.0014)		(0.0014)
Fit statistics										
Observations	597	597	597	597	597	597	597	597	597	597
R^2	0.01060	0.01452	0.00600	0.00710	0.01105	0.00695	0.01107	0.00606	0.00899	0.00650
Adjusted R^2	0.00392	0.00954	-0.00072	0.00208	0.00437	0.00193	0.00438	0.00103	0.00229	0.00147

IID standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: This table presents five regressions using the form presented in Equation 3. In each model, an individual's sensitivity of a belief to consensus, captured by the slope of the individual-level regression of the belief on consensus, is regressed on a constant and either demographic or prior belief attributes. The dependent variables are grouped for each second column and are, in order, belief in human-caused climate change, the skill of pro-consensus scientists, the skill of anti-consensus scientists, the bias of pro-consensus scientists, and the bias of anti-consensus scientists. In the odd columns, the independent variables are demographics including a continuous measure of age, and dummy variables for if the participant is female, completed college, and is white. The even columns use continuous measures of prior beliefs as independent variables including belief in human-caused climate change, that a randomly selected climate scientist will be unbiased, and that a randomly selected climate scientist will be skilled. All models use classical standard errors.

S1.2 Structural Equation Models and Robustness

To check the robustness of the structural equation models, we use multiple regression models. The first models are precisely the population-level version of Equation 1 whose results have been presented in Sections 11.1 and 11.2. The second model is a regression of belief in human-caused climate change on beliefs about the attributes of scientists and a continuous variable for consensus. These results are presented in Table S5.

Table S5: OLS model of relationship between scientist attributes and climate belief

Dependent Variable:		Climate	
Model:	(1)	(2)	(3)
Variables			
Constant	5.716*		
	(3.102)		
ProSkill	0.6166***	0.5263***	0.5262***
	(0.0374)	(0.0553)	(0.0554)
AntiSkill	-0.0516*	0.1274***	0.1290***
	(0.0289)	(0.0469)	(0.0469)
ProBias	0.0308	0.1395***	0.1393***
	(0.0290)	(0.0481)	(0.0479)
AntiBias	0.1217***	0.0940**	0.0944**
	(0.0308)	(0.0437)	(0.0437)
Consensus	0.1744***	0.1610***	
	(0.0250)	(0.0255)	
Fixed-effects			
ID	No	Yes	Yes
Consensus	No	No	Yes
Fit statistics			
Observations	2,985	2,985	2,985
R^2	0.42821	0.83217	0.83270
Within R^2		0.40257	0.32260

Clustered (ID) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: This table presents three regressions using the form presented in Equation 1 with analyses outlined in Methods Subsection 3.3. An individual's belief in human-caused climate change at a consensus level is regressed on their other beliefs namely in the skill of pro-consensus scientists (ProSkill), the skill of anti-consensus scientists (AntiSkill), the bias of pro-consensus scientists (ProBias), and the bias of anti-consensus scientists (AntiBias). In columns (1) and (2) a continous measure of consensus is included. Meanwhile in columns (2) and (3) individual fixed-effects are inclued. In column (3) the continuous measure of consensus is replaced by a fixed effect for consensus. All standard errors are clustered at the individual-level.

S1.3 Samples and Demographic Information

Table S6: Demographic Summary Statistics of Bovitz Sample

Variable	N	Mean	SD	National
Age	597	48.26	16.91	41.7
Female	597	0.49	0.5	0.51
White	597	0.75	0.43	0.76
College	597	0.37	0.48	0.34
Income	390			$69k
... [$0, $40k)	93	24%		
... [$40k, $80k)	198	51%		
... [$80k, $150k)	65	17%		
... $\geq$ 150k	34	9%		
Party	597			
... Democrat	205	34%		29%
... Independent	178	30%		35%
... Other (specify)	17	3%		
... Republican	197	33%		33%

Notes: This table presents demographic summary statistics for the collected demographics of the Bovitz Sample. Age is a continuous variable, Female is a dummy variable for if the individual identifies as female, White is a dummy variable for if the individual reports being white, Income is a categorical presenting buckets for household income, while Party is a categorical variable for self-reported political party identification. The "National" column presents the national average for that demographic, with the exception of age and household income which are medians. Female, White, College, and Income are taken from the US Census Bureau Quick Stats. Median age is the median age of the labor force (the relevant population for online surveys) and is taken from the Bureau of Labor Statistics. The mean age in our sample should be expected to differ from the median age in the population. Political party affiliation is taken from an October 2022 Gallup survey.

S2 Supplementary Information II: MTurk Sample

This section of the supplementary information presents results using the within-subject experiment using the MTurk sample.

S2.1 Replication of Main Results with MTurk Data

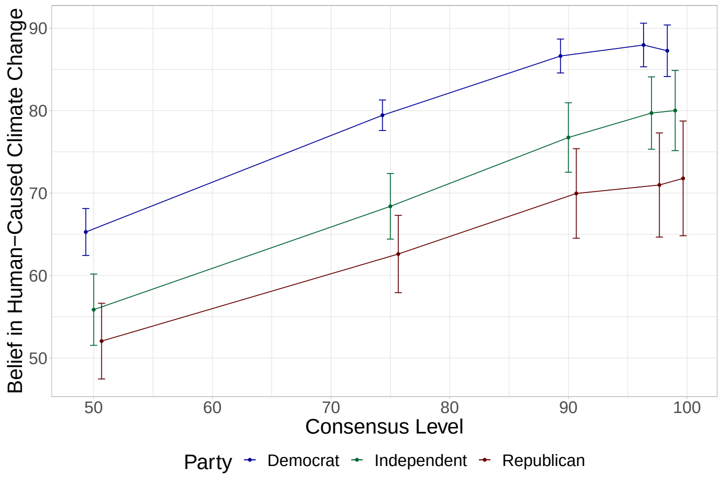

Fig. S1: **The Responsiveness of Belief in Human-Caused Climate Change to Scientific Consensus, Split by Political Party.** The mean belief is plotted at 50%, 75%, 90%, 97%, and 99% consensus with 95% confidence intervals. This is a replication of Figure 2 with MTurk data.

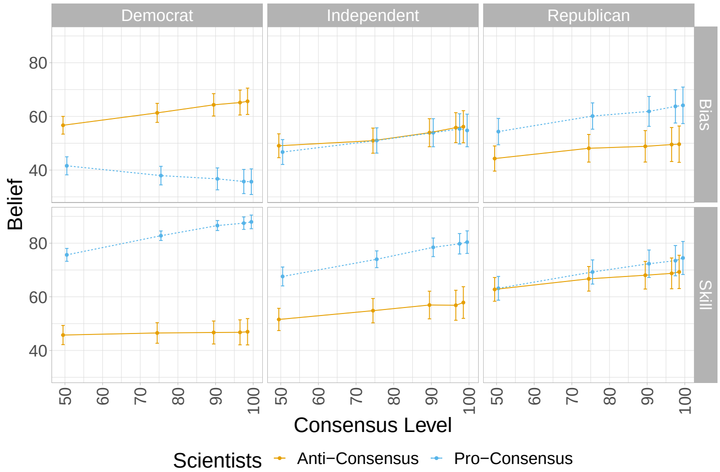

Fig. S2: **The Responsiveness of Beliefs in About the Attributes of Climate Scientists to Scientific Consensus, Split by Political Party.** The mean belief is plotted at 50%, 75%, 90%, 97%, and 99% consensus with 95% confidence intervals. This is a replication of Figure 3 with MTurk data.

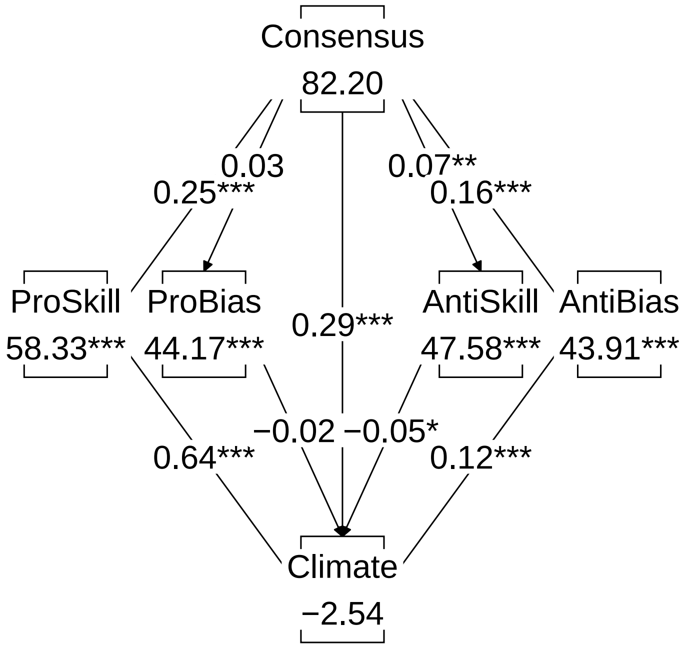

Fig. S3: **Structural Equation Model Demonstrating Direct and Indirect Pathways between Consensus and Belief in Human-Caused Climate Change.** Beliefs are expressed in linear space and standard errors are clustered by individual. This is a replication of Figure 4 with MTurk data.

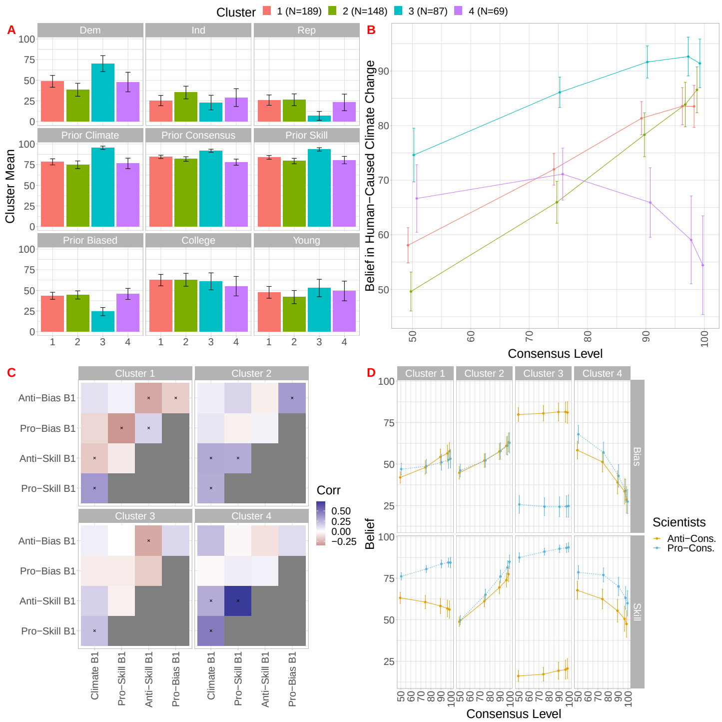

Fig. S4: **The Population can be Decomposed into 4 Distinct Clusters that have Different Belief Patterns and Correspond to Different Demographics and Prior Beliefs.** Panel A shows the percent of the cluster that is Democrat, Independent, and Republican, attended college, has a below median age (Young), and their corresponding average prior beliefs. Panel B shows the average belief in human-caused climate change at each consensus level split by cluster. Panel C shows the correlation between the belief slopes within each cluster. Statistically significant correlations are marked with an "X". Finally, Panel D shows the average belief in each of the 4 scientist attributes split by cluster. 95% confidence intervals are shown around all means. This is a replication of Figure 5 with MTurk data.

S2.2 Replication of Supplementary Results with MTurk Data

Table S7: Summary Statistics of Individual-Level Sensitivity to Consensus in MTurk Data

Variable	N	Mean	SD	Pct ¿ 0	T-stat
Climate Slope	493	0.46	0.72	0.82	t = 14.29
Pro-Skill Slope	493	0.25	0.53	0.78	t = 10.50
Anti-Skill Slope	493	0.07	0.60	0.57	t = 2.74
Pro-Bias Slope	493	0.04	0.72	0.54	t = 1.04
Anti-Bias Slope	493	0.16	0.64	0.65	t = 5.37

Notes: This is equivalent to Table S1 with MTurk data and all table notes are identical. The mean and standard deviation of the distribution of sensitivity to consensus is shown for each belief type. Sensitivity to consensus is measured using an individual-level linear regression of belief on consensus level. The column titled $Pct > 0$ provides the proportion of the distribution that has a positive slope. Finally, a t-stat from a t-test of each distribution against the null-hypothesis that the mean of the distribution is 0 is presented.

Table S8: Regression of Individual Beliefs on Scientist Type and Party Identification

Dependent Variables:	Skill		Bias	
Model:	(1)	(2)	(3)	(4)
Variables				
Pro	25.07***	3.435	-9.760***	12.76***
	(1.683)	(3.159)	(1.987)	(4.011)
Dem		-20.56***		14.55***
		(3.208)		(3.352)
Ind		-11.50***		5.101
		(3.449)		(3.669)
Pro × Dem		34.10***		-37.86***
		(3.876)		(4.813)
Pro × Ind		17.00***		-13.58***
		(4.363)		(5.240)
Fixed-effects				
ID	Yes	No	Yes	No
Consensus	Yes	Yes	Yes	Yes
Fit statistics				
Observations	4,930	4,930	4,930	4,930
R^2	0.50228	0.22525	0.40330	0.08317
Within R^2	0.24894	0.21813	0.03444	0.08073

Clustered (ID) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: This is a replication of Table S2 with MTurk data. The following table notes are identical. This table presents four regressions using the form presented in Equation 2. Columns (1) and (2) regress an individual's belief in the skill of scientists at a given consensus level on a dummy for if the scientist is pro-consensus and, in Column (2), dummies for political party identification and their interaction with *Pro*. Columns (3) and (4) use the same models with scientist bias as the dependent variable. In all columns, there are consensus-level fixed-effects while columns (1) and (3) also include individual fixed-effects. In all models, standard errors are clustered by individual.

Table S9: Regression of Individual Sensitivity of Beliefs to Consensus Level on Party Identification with MTurk Data

Dependent Variables:	Climate	ProSkill	AntiSKill	ProBias	AntiBias
Model:	(1)	(2)	(3)	(4)	(5)
Variables					
Constant	0.5037***	0.2628***	0.1227**	0.1746***	0.1474***
	(0.0610)	(0.0445)	(0.0510)	(0.0599)	(0.0545)
Dem	-0.0375	-0.0114	-0.1000	-0.2945***	0.0352
	(0.0766)	(0.0559)	(0.0640)	(0.0752)	(0.0684)
Rep	-0.0945	-0.0386	0.0049	0.0195	-0.0412
	(0.0920)	(0.0671)	(0.0769)	(0.0903)	(0.0821)
Fit statistics					
Observations	493	493	493	493	493
R^2	0.00216	0.00070	0.00716	0.04401	0.00223
Adjusted R^2	-0.00191	-0.00338	0.00311	0.04011	-0.00184

IID standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: This table is equivalent to Table S3 with MTurk data. All table notes are identical. The table presents five regressions using the form presented in Equation 3. In each model, an individual's sensitivity of a belief to consensus, captured by the slope of the individual-level regression of the belief on consensus, is regressed on a constant and their party identification. The dependent variables are, in order, belief in human-caused climate change, the skill of pro-consensus scientists, the skill of anti-consensus scientists, the bias of pro-consensus scientists, and the bias of anti-consensus scientists. Classical standard errors are used in each model.

Table S10: OLS model of relationship between scientist attributes and climate belief with MTurk data

Dependent Variable:		Climate	
Model:	(1)	(2)	(3)
Variables			
Constant	-2.537		
	(3.323)		
ProSkill	0.6393***	0.5648***	0.5626***
	(0.0459)	(0.0604)	(0.0602)
AntiSkill	-0.0495**	0.0338	0.0344
	(0.0208)	(0.0454)	(0.0453)
ProBias	-0.0194	0.0579	0.0579
	(0.0217)	(0.0482)	(0.0483)
AntiBias	0.1248***	0.1647***	0.1654***
	(0.0240)	(0.0492)	(0.0493)
Consensus	0.2901***	0.2936***	
	(0.0319)	(0.0335)	
Fixed-effects			
ID	No	Yes	Yes
Consensus	No	No	Yes
Fit statistics			
Observations	2,465	2,465	2,465
R^2	0.46566	0.77593	0.77678
Within R^2		0.43557	0.24727

Clustered (ID) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: This table is equivalent to Table S5 with MTurk data and all table notes are identical. The table presents three regressions using the form presented in Equation 1 with analyses outlined in Methods Subsection 3.3. An individual's belief in human-caused climate change at a consensus level is regressed on their other beliefs namely in the skill of pro-consensus scientists (ProSkill), the skill of anti-consensus scientists (AntiSkill), the bias of pro-consensus scientists (ProBias), and the bias of anti-consensus scientists (AntiBias). In columns (1) and (2) a continous measure of consensus is included. Meanwhile in columns (2) and (3) individual fixed-effects are inclued. In column (3) the continuous measure of consensus is replaced by a fixed effect for consensus. All standard errors are clustered at the individual-level.

Table S11: Demographic Summary Statistics of MTurk Sample

Variable	N	Mean	SD
Age	493	41.11	12.53
Female	493	0.52	0.5
White	493	0.83	0.38
College	493	0.61	0.49
Income	345		
... [$0, $40k)	56	16%	
... [$40k, $80k)	200	58%	
... [$80k, $150k)	63	18%	
... geq 150k	26	8%	
Party	493		
... Democrat	243	49%	
... Independent	130	26%	
... Other (specify)	10	2%	
... Republican	110	22%	

Notes: This table is equivalent to Table S6 and all table notes are identical. The table presents demographic summary statistics for the collected demographics of the Bovitz Sample. Age is a continous variable, Female is a dummy variable for if the individual identifies as female, White is a dummy variable for if the individual reports being white, Income is a categorical presenting buckets for household income, while Party is a categorical variable for self-reported political party identification.

S3 Supplementary Information III: Between Subject Replication

This section presents results from a between-subject version of the experiment conducted on MTurk.

S3.1 Between Subject Demographics

Table S12: Demographic Summary Statistics of Between Subject MTurk Sample

Variable	N	Mean	SD
Age	432	40.63	12.6
Female	432	0.45	0.5
White	432	0.78	0.42
College	432	0.61	0.49
Party	432		
... Democrat	219	51%	
... Independent	100	23%	
... Other (specify)	5	1%	
... Republican	108	25%	

Notes: This table presents demographic summary statistics for the collected demographics of the Bovitz Sample. Age is a continuous variable, Female is a dummy variable for if the individual identifies as female, White is a dummy variable for if the individual reports being white, while Party is a categorical variable for self-reported political party identification.

S3.2 Between Subject Beliefs

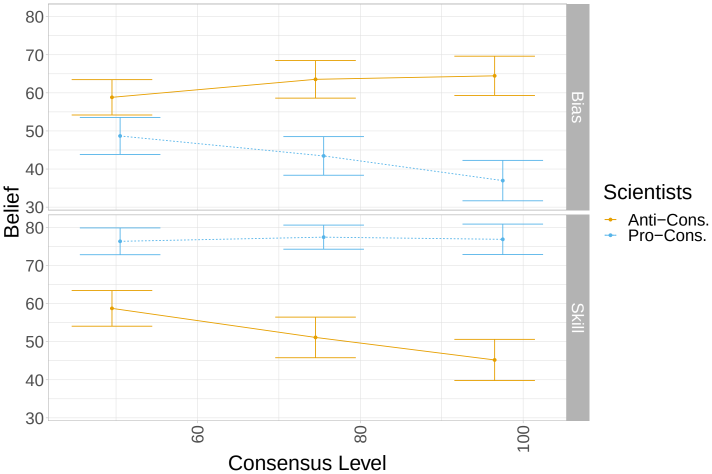

Fig. S5: **The Between-Subject Responsiveness of Beliefs About the Attributes of Climate Scientists to Scientific Consensus.** The mean belief is plotted at 50%, 75%, and 97% consensus with 95% confidence intervals.

S4 Supplementary Information IV: Intervention Pre-tests

We conducted pre-tests with 49 mTurk participants to check that our proposed interventions had the anticipated effect on beliefs about scientist attributes. We asked the participants to rate how each intervention changed their beliefs on three hypotheses: 1) the skill of climate scientists, 2) the honesty of climate scientists, and 3) the likelihood that human-caused climate change is occurring. Participants rated these changes on a five-point Likert scale. Figure S6 shows the mean change in each of the three beliefs for the two interventions. As predicted, interventions had positive effects on beliefs.

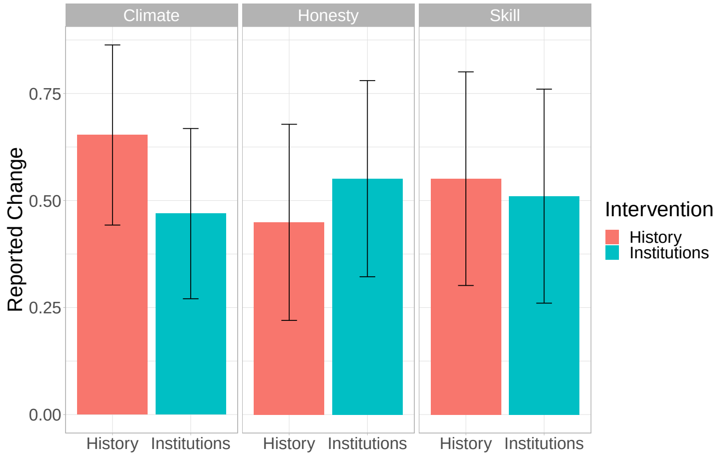

Fig. S6: **Mean Reported Changes in Focal Beliefs by Intervention Type in Pretests.** The change in belief was originally measured on a five-point Likert scale. Here, we normalize the scale such that the midpoint (no change) is 0. The mean change in belief is shown with 95% confidence intervals. The top facets correspond to the three beliefs measured—human-caused climate change, the honesty of climate scientists, and the skill of climate scientists. The orange bar shows the "History" intervention while the blue bar shows the "Institutions" intervention.

S5 Supplementary Information V: Intervention Robustness

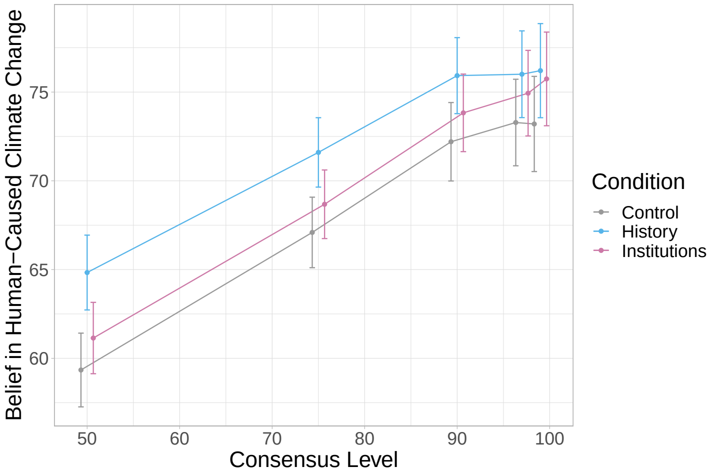

Fig. S7: **Relationship between Consensus and Climate Belief by Experimental Condition.** 95% confidence intervals are shown around all means.

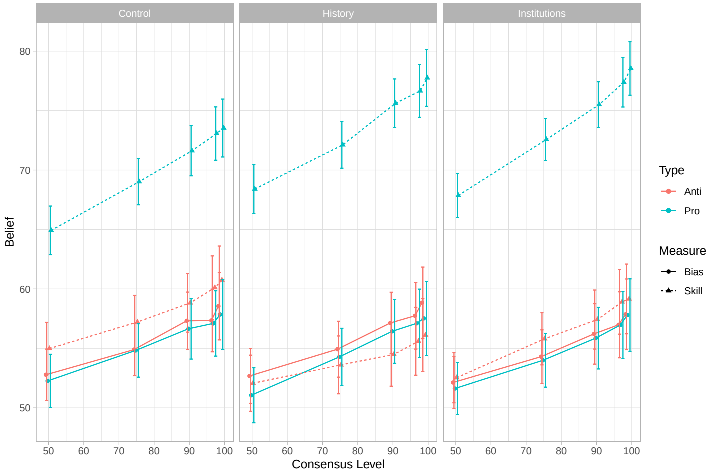

Fig. S8: **Relationship between Consensus and Scientist Attributes by Experimental Condition.** 95% confidence intervals are shown around all means.

S6 Supplementary Information VI: Hierarchical Bayesian Model

Below we present a hierarchical Bayesian model that qualitatively captures the patterns of joint inference observed in the experimental results. A directed acyclic graph (DAG) summarizing the model is presented in Figure S9. The nodes inside the plate are repeated for every scientist in the set of scientists (in this case 100).

Intuitively, an individual scientist i reports whether or not they believe human-caused climate change is occurring C_i. Whether they report it is occurring is a function of whether it is actually occurring (represented by H), their skill (S_i), and their bias (T_i). A scientist can be unbiased ($T_i = 0$) or biased in the direction of saying climate change is ($T_i = 1$) or is not ($T_i = -1$) occurring. When a scientist is unbiased, they will use their skill to determine whether or not climate change is happening. If the scientist is skilled, they are more likely to correctly match C_i to the true value of H_i. Meanwhile, if a scientist is biased, they are disproportionately more likely to say that climate change is or is not occurring due to that bias.

The key feature of this together. In this case, K represents whether or not the field, in general, is skilled and the domain is knowable while $B_i \in \{-1, 0, 1\}$ provides the directional bias of the field.

With this model, we can query the relevant posterior beliefs that we elicit in the survey. The first relevant query is about human-caused climate change given a certain level of scientific consensus, $P(H|C = c)$. We can also query beliefs about the attributes of scientists given a certain consensus level, C, *and* the expressed position of the scientist. The bias of a pro-consensus scientist is given by $P(T_i = 1|C_i = 1, C = c)$ while the bias of an anti-consensus scientist is $P(T_i = -1|C_i = 1, C = c)$. Finally, we can query the probability that a pro-consensus or anti-consensus scientist is skilled given a consensus level and that they are unbiased, $P(S_i = 1|C_i = 1, T_i = 0, C = c)$ and $P(S_i = 0|C_i = 0, T_i = 0, C = c)$ respectively. Importantly, the skill and bias of all pro- or anti-consensus scientists are identical so we only need to query the model's belief about one scientist in the group.

We ran simulations over a grid of prior beliefs and likelihoods to capture the behavior of the model. We performed model iterations, with a query at each consensus level, for the combination of the following parameter values:

$$
\begin{aligned}
P(H = 1) &\in \{0.1, 0.3, 0.5, 0.7, 0.9\} \\
P(K = 1) &\in \{0.1, 0.3, 0.5, 0.7, 0.9\} \\
P(B = 1) = P(B = -1) &\in \{0.05, 0.15, 0.25, 0.35, 0.45\} \\
P(S_i = k|K = k) &\in \{0.9\} \\
P(T_i = b|B = b) &\in \{0.9\} \\
P(C_i = h|S_i = 1, T_i = 0, H = h) &\in \{0.7, 0.9\} \\
P(C_i = 1|T_i = 1) = P(C_i = 0|T_i = -1) &\in \{0.7, 0.9\}
\end{aligned}
$$

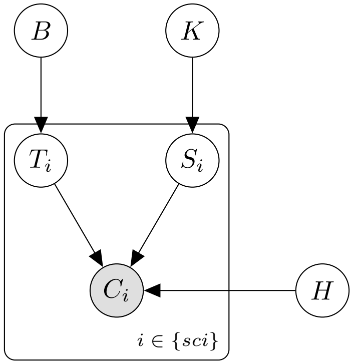

Fig. S9: **Hierarchical Bayesian Model of Expert Consensus**

where $P(S_i = k|K = k)$ is the likelihood that scientist i is skilled or not given that the field is knowable/generally skilled or not, $P(T_i = b|B = b)$ is the likelihood that scientist i is biased in direction b (including unbiased when $b = 0$) given that the field is biased in direction b. Meanwhile, $P(C_i = h|S_i = 1, T_i = 0, H = h)$ gives the likelihood that scientist i correctly states the true state of climate change h given that they are skilled and unbiased. $P(C_i = 1|T_i = 1)$ and $P(C_i = 0|T_i = -1)$ given the probability that a directionally biased climate scientist will say that climate change is or is not occurring in that direction irrespective of their skill and the state of the world. These parameter values were picked such that the likelihoods were greater than 0.5, where the intuition of the model would fall apart (for example, a skilled scientist being worse than chance at identifying if climate change is occurring). The prior probability of bias was modeled such that pro- and anti-consensus scientists were ex-ante equally likely to be biased. This means that for $P(B = 1) = P(B = -1) = 0.45$ the overall probability of bias in the field is 0.9.

We present iterations of the model for one combination of likelihoods as an example below. The results correspond to the set of likelihoods: $P(S_i = k|K = k) = 0.9$; $P(T_i = b|B = b) = 0.7$; $P(C_i = 1|T_i = 1) = P(C_i = 0|T_i = -1) = 0.9$; $P(C_i = h|S_i = 1, T_i = 0, H = h) = 0.7$. We are able to capture the qualitative patterns of the empirical results with belief in human-caused climate change and all scientist attributes increasing with consensus level.

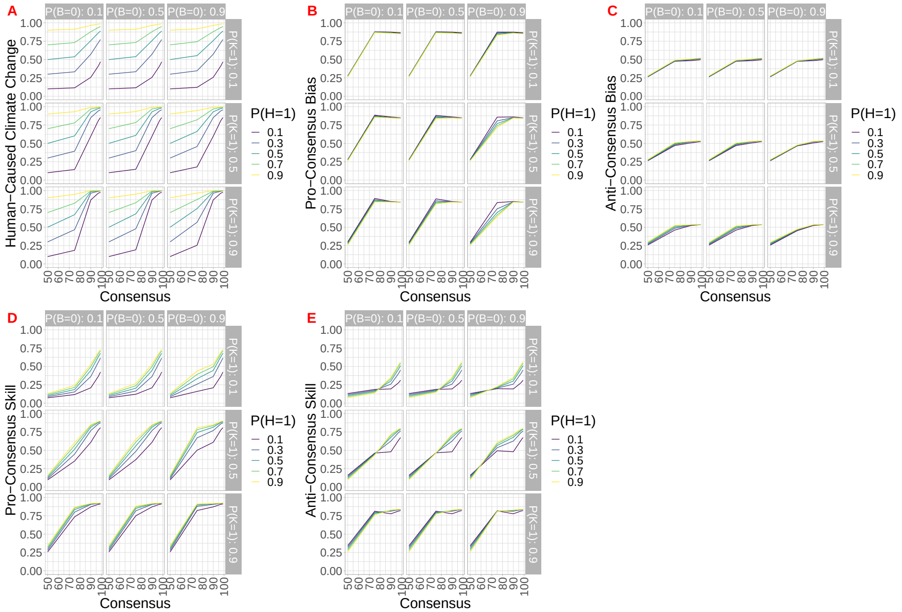

Fig. S10: **Simulations from the Hierarchical Bayesian Model with a Qualitative Match to Posterior Belief Patterns in Main Results.** We estimate posterior beliefs for each of the 5 consensus levels in the survey: 50%, 75%, 90%, 97%, 99%. The top facets labeled $P(B = 0)$ give the prior belief that the field is unbiased, the right facets labeled $P(K = 1)$ give the prior belief that the field is skilled/the domain is knowable, while the line color gives the prior belief in human-caused climate change $P(H = 1)$. Panel A shows the posterior belief in human-caused climate change given consensus, $P(H|C = c)$. Panel B shows the posterior belief that a pro-consensus scientist will be biased to say that climate change is occurring, $P(T_i = 1|C_i = 1, C = c)$. Panel C shows the posterior belief that an anti-consensus scientist will be biased against saying that climate change is occurring, $P(T_i = -1|C_i = 0, C = c)$. Panel D gives the posterior belief that a pro-consensus scientist is skilled, $P(S_i = 1|C_i = 1, T_i = 0, C = c)$, while Panel E shows the belief that a anti-consensus scientists is skilled, $P(S_i = 1|C_i = 0, T_i = 0, C = c)$.

The remaining simulation iterations are presented below split by the combination of likelihoods used to produce them.

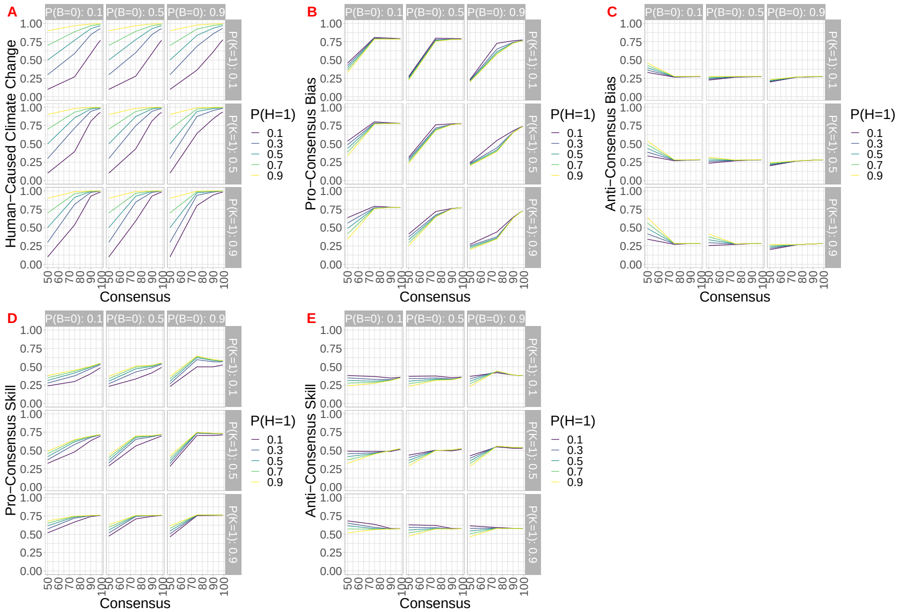

Fig. S11: **Simulation from the Hierarchical Bayesian Model with Likelihoods:** $P(S_i = k|K = k) = 0.7$; $P(T_i = b|B = b) = 0.7$; $P(C_i = 1|T_i = 1) = P(C_i = 0|T_i = -1) = 0.7$; $P(C_i = h|S_i = 1, T_i = 0, H = h) = 0.7$ We estimate posterior beliefs for each of the 5 consensus levels in the survey: 50%, 75%, 90%, 97%, 99%. The top facets labeled $P(B = 0)$ give the prior belief that the field is unbiased, the right facets labeled $P(K = 1)$ give the prior belief that the field is skilled/the domain is knowable, while the line color gives the prior belief in human-caused climate change $P(H = 1)$. Panel A shows the posterior belief in human-caused climate change given consensus, $P(H|C = c)$. Panel B shows the posterior belief that a pro-consensus scientist will be biased to say that climate change is occurring, $P(T_i = 1|C_i = 1, C = c)$. Panel C shows the posterior belief that an anti-consensus scientist will be biased against saying that climate change is occurring, $P(T_i = -1|C_i = 0, C = c)$. Panel D gives the posterior belief that a pro-consensus scientist is skilled, $P(S_i = 1|C_i = 1, T_i = 0, C = c)$, while Panel E shows the belief that a anti-consensus scientists is skilled, $P(S_i = 1|C_i = 0, T_i = 0, C = c)$.

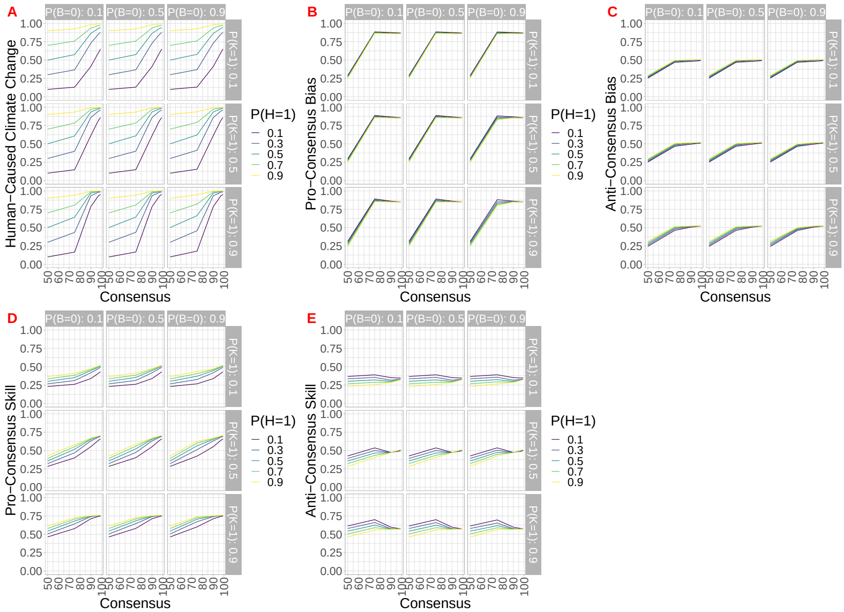

Fig. S12: **Simulation from the Hierarchical Bayesian Model with Likelihoods:** $P(S_i = k|K = k) = 0.7$; $P(T_i = b|B = b) = 0.7$; $P(C_i = 1|T_i = 1) = P(C_i = 0|T_i = -1) = 0.9$; $P(C_i = h|S_i = 1, T_i = 0, H = h) = 0.7$ We estimate posterior beliefs for each of the 5 consensus levels in the survey: 50%, 75%, 90%, 97%, 99%. The top facets labeled $P(B = 0)$ give the prior belief that the field is unbiased, the right facets labeled $P(K = 1)$ give the prior belief that the field is skilled/the domain is knowable, while the line color gives the prior belief in human-caused climate change $P(H = 1)$. Panel A shows the posterior belief in human-caused climate change given consensus, $P(H|C = c)$. Panel B shows the posterior belief that a pro-consensus scientist will be biased to say that climate change is occurring, $P(T_i = 1|C_i = 1, C = c)$. Panel C shows the posterior belief that an anti-consensus scientist will be biased against saying that climate change is occurring, $P(T_i = -1|C_i = 0, C = c)$. Panel D gives the posterior belief that a pro-consensus scientist is skilled, $P(S_i = 1|C_i = 1, T_i = 0, C = c)$, while Panel E shows the belief that a anti-consensus scientists is skilled, $P(S_i = 1|C_i = 0, T_i = 0, C = c)$.

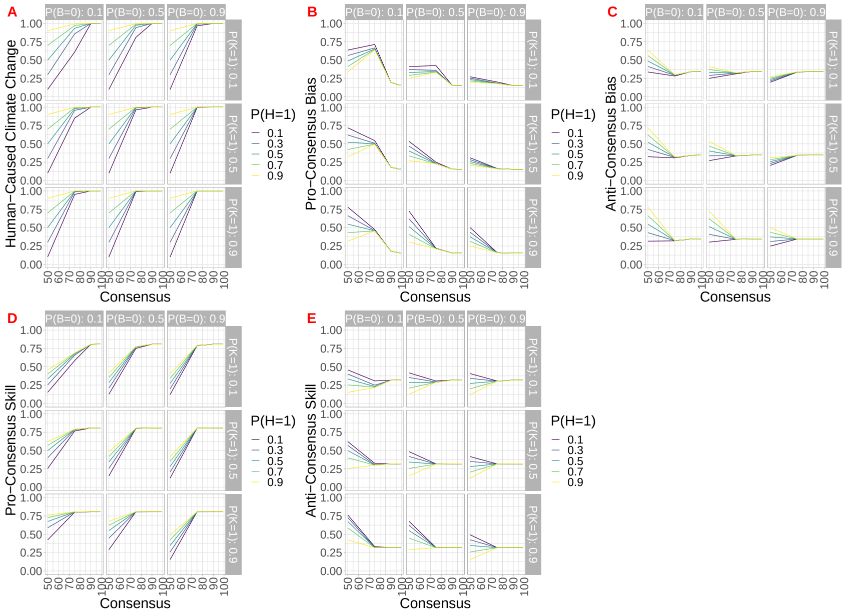

Fig. S13: **Simulation from the Hierarchical Bayesian Model with Likelihoods:** $P(S_i = k|K = k) = 0.7$; $P(T_i = b|B = b) = 0.7$; $P(C_i = 1|T_i = 1) = P(C_i = 0|T_i = -1) = 0.7$; $P(C_i = h|S_i = 1, T_i = 0, H = h) = 0.9$ We estimate posterior beliefs for each of the 5 consensus levels in the survey: 50%, 75%, 90%, 97%, 99%. The top facets labeled $P(B = 0)$ give the prior belief that the field is unbiased, the right facets labeled $P(K = 1)$ give the prior belief that the field is skilled/the domain is knowable, while the line color gives the prior belief in human-caused climate change $P(H = 1)$. Panel A shows the posterior belief in human-caused climate change given consensus, $P(H|C = c)$. Panel B shows the posterior belief that a pro-consensus scientist will be biased to say that climate change is occurring, $P(T_i = 1|C_i = 1, C = c)$. Panel C shows the posterior belief that an anti-consensus scientist will be biased against saying that climate change is occurring, $P(T_i = -1|C_i = 0, C = c)$. Panel D gives the posterior belief that a pro-consensus scientist is skilled, $P(S_i = 1|C_i = 1, T_i = 0, C = c)$, while Panel E shows the belief that a anti-consensus scientists is skilled, $P(S_i = 1|C_i = 0, T_i = 0, C = c)$.

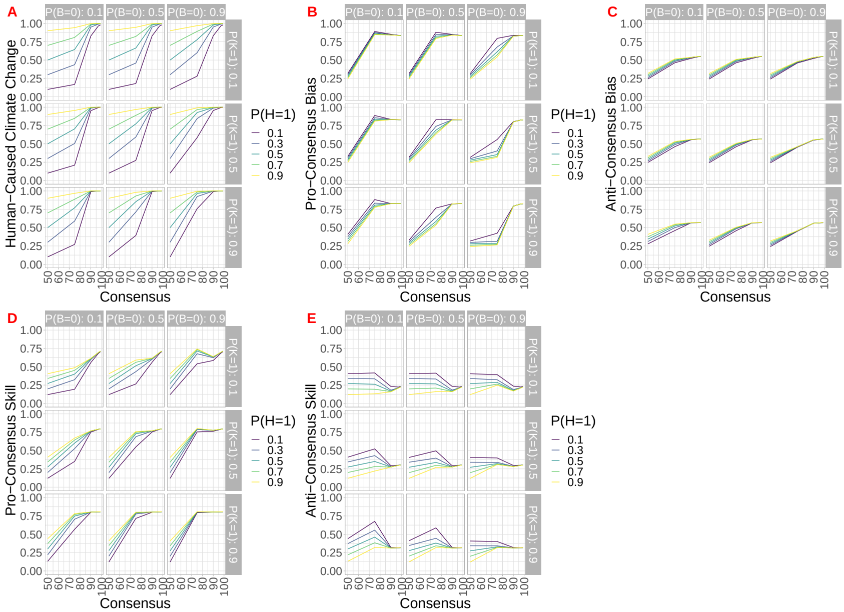

Fig. S14: **Simulation from the Hierarchical Bayesian Model with Likelihoods:** $P(S_i = k|K = k) = 0.7$; $P(T_i = b|B = b) = 0.7$; $P(C_i = 1|T_i = 1) = P(C_i = 0|T_i = -1) = 0.9$; $P(C_i = h|S_i = 1, T_i = 0, H = h) = 0.9$ We estimate posterior beliefs for each of the 5 consensus levels in the survey: 50%, 75%, 90%, 97%, 99%. The top facets labeled $P(B = 0)$ give the prior belief that the field is unbiased, the right facets labeled $P(K = 1)$ give the prior belief that the field is skilled/the domain is knowable, while the line color gives the prior belief in human-caused climate change $P(H = 1)$. Panel A shows the posterior belief in human-caused climate change given consensus, $P(H|C = c)$. Panel B shows the posterior belief that a pro-consensus scientist will be biased to say that climate change is occurring, $P(T_i = 1|C_i = 1, C = c)$. Panel C shows the posterior belief that an anti-consensus scientist will be biased against saying that climate change is occurring, $P(T_i = -1|C_i = 0, C = c)$. Panel D gives the posterior belief that a pro-consensus scientist is skilled, $P(S_i = 1|C_i = 1, T_i = 0, C = c)$, while Panel E shows the belief that a anti-consensus scientists is skilled, $P(S_i = 1|C_i = 0, T_i = 0, C = c)$.

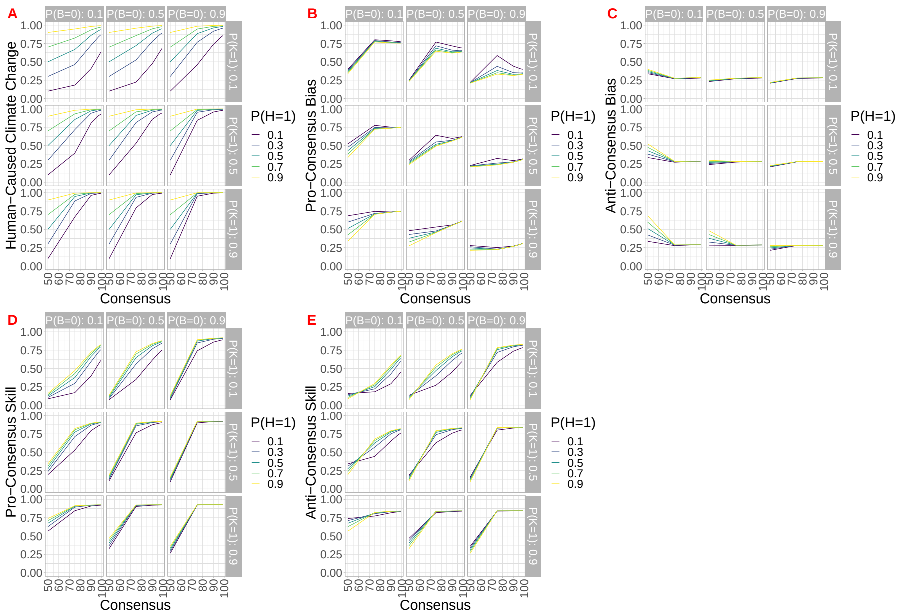

Fig. S15: **Simulation from the Hierarchical Bayesian Model with Likelihoods:** $P(S_i = k|K = k) = 0.9$; $P(T_i = b|B = b) = 0.7$; $P(C_i = 1|T_i = 1) = P(C_i = 0|T_i = -1) = 0.7$; $P(C_i = h|S_i = 1, T_i = 0, H = h) = 0.7$ We estimate posterior beliefs for each of the 5 consensus levels in the survey: 50%, 75%, 90%, 97%, 99%. The top facets labeled $P(B = 0)$ give the prior belief that the field is unbiased, the right facets labeled $P(K = 1)$ give the prior belief that the field is skilled/the domain is knowable, while the line color gives the prior belief in human-caused climate change $P(H = 1)$. Panel A shows the posterior belief in human-caused climate change given consensus, $P(H|C = c)$. Panel B shows the posterior belief that a pro-consensus scientist will be biased to say that climate change is occurring, $P(T_i = 1|C_i = 1, C = c)$. Panel C shows the posterior belief that an anti-consensus scientist will be biased against saying that climate change is occurring, $P(T_i = -1|C_i = 0, C = c)$. Panel D gives the posterior belief that a pro-consensus scientist is skilled, $P(S_i = 1|C_i = 1, T_i = 0, C = c)$, while Panel E shows the belief that a anti-consensus scientists is skilled, $P(S_i = 1|C_i = 0, T_i = 0, C = c)$.

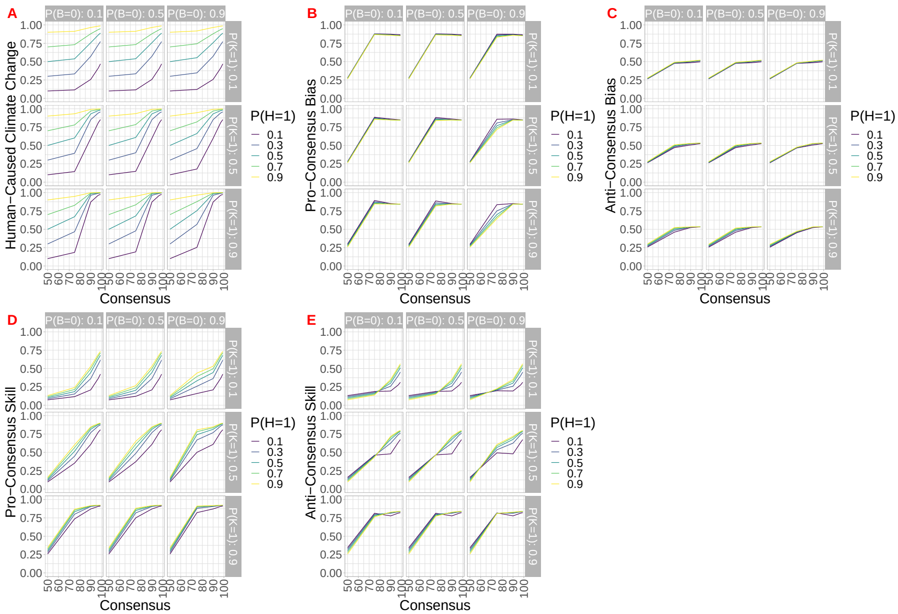

Fig. S16: **Simulation from the Hierarchical Bayesian Model with Likelihoods:** $P(S_i = k|K = k) = 0.9$; $P(T_i = b|B = b) = 0.7$; $P(C_i = 1|T_i = 1) = P(C_i = 0|T_i = -1) = 0.9$; $P(C_i = h|S_i = 1, T_i = 0, H = h) = 0.7$ We estimate posterior beliefs for each of the 5 consensus levels in the survey: 50%, 75%, 90%, 97%, 99%. The top facets labeled $P(B = 0)$ give the prior belief that the field is unbiased, the right facets labeled $P(K = 1)$ give the prior belief that the field is skilled/the domain is knowable, while the line color gives the prior belief in human-caused climate change $P(H = 1)$. Panel A shows the posterior belief in human-caused climate change given consensus, $P(H|C = c)$. Panel B shows the posterior belief that a pro-consensus scientist will be biased to say that climate change is occurring, $P(T_i = 1|C_i = 1, C = c)$. Panel C shows the posterior belief that an anti-consensus scientist will be biased against saying that climate change is occurring, $P(T_i = -1|C_i = 0, C = c)$. Panel D gives the posterior belief that a pro-consensus scientist is skilled, $P(S_i = 1|C_i = 1, T_i = 0, C = c)$, while Panel E shows the belief that a anti-consensus scientists is skilled, $P(S_i = 1|C_i = 0, T_i = 0, C = c)$.

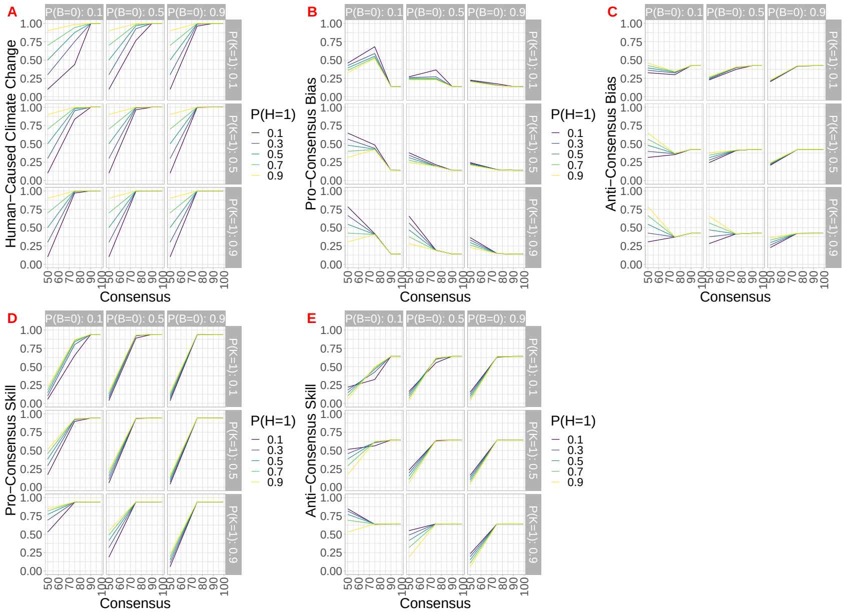

Fig. S17: **Simulation from the Hierarchical Bayesian Model with Likelihoods:** $P(S_i = k|K = k) = 0.9$; $P(T_i = b|B = b) = 0.7$; $P(C_i = 1|T_i = 1) = P(C_i = 0|T_i = -1) = 0.7$; $P(C_i = h|S_i = 1, T_i = 0, H = h) = 0.9$ We estimate posterior beliefs for each of the 5 consensus levels in the survey: 50%, 75%, 90%, 97%, 99%. The top facets labeled $P(B = 0)$ give the prior belief that the field is unbiased, the right facets labeled $P(K = 1)$ give the prior belief that the field is skilled/the domain is knowable, while the line color gives the prior belief in human-caused climate change $P(H = 1)$. Panel A shows the posterior belief in human-caused climate change given consensus, $P(H|C = c)$. Panel B shows the posterior belief that a pro-consensus scientist will be biased to say that climate change is occurring, $P(T_i = 1|C_i = 1, C = c)$. Panel C shows the posterior belief that an anti-consensus scientist will be biased against saying that climate change is occurring, $P(T_i = -1|C_i = 0, C = c)$. Panel D gives the posterior belief that a pro-consensus scientist is skilled, $P(S_i = 1|C_i = 1, T_i = 0, C = c)$, while Panel E shows the belief that a anti-consensus scientists is skilled, $P(S_i = 1|C_i = 0, T_i = 0, C = c)$.

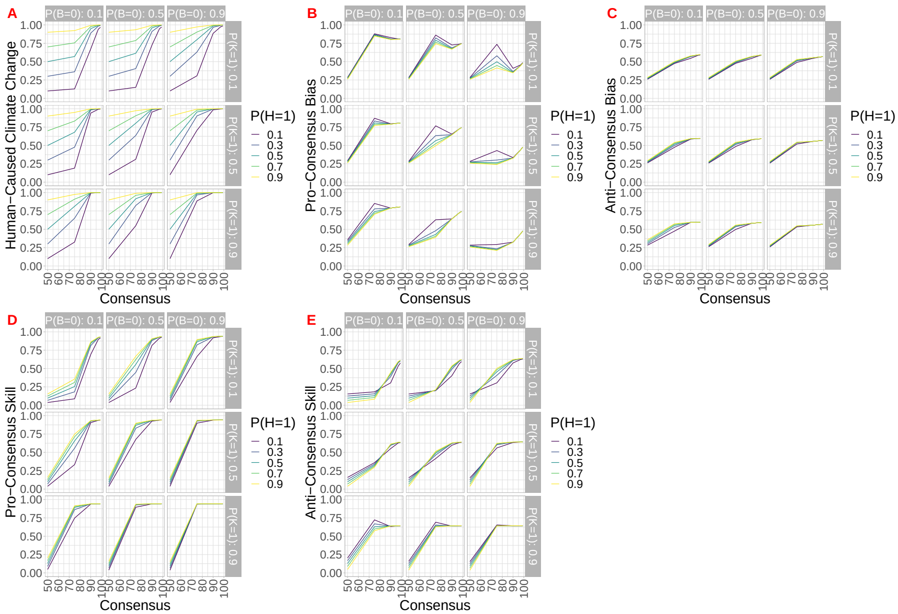

Fig. S18: **Simulation from the Hierarchical Bayesian Model with Likelihoods:** $P(S_i = k|K = k) = 0.9$; $P(T_i = b|B = b) = 0.7$; $P(C_i = 1|T_i = 1) = P(C_i = 0|T_i = -1) = 0.9$; $P(C_i = h|S_i = 1, T_i = 0, H = h) = 0.9$ We estimate posterior beliefs for each of the 5 consensus levels in the survey: 50%, 75%, 90%, 97%, 99%. The top facets labeled $P(B = 0)$ give the prior belief that the field is unbiased, the right facets labeled $P(K = 1)$ give the prior belief that the field is skilled/the domain is knowable, while the line color gives the prior belief in human-caused climate change $P(H = 1)$. Panel A shows the posterior belief in human-caused climate change given consensus, $P(H|C = c)$. Panel B shows the posterior belief that a pro-consensus scientist will be biased to say that climate change is occurring, $P(T_i = 1|C_i = 1, C = c)$. Panel C shows the posterior belief that an anti-consensus scientist will be biased against saying that climate change is occurring, $P(T_i = -1|C_i = 0, C = c)$. Panel D gives the posterior belief that a pro-consensus scientist is skilled, $P(S_i = 1|C_i = 1, T_i = 0, C = c)$, while Panel E shows the belief that a anti-consensus scientists is skilled, $P(S_i = 1|C_i = 0, T_i = 0, C = c)$.